DP Computer Science: A 2.1.4 Network Protocols

1. Core Concepts & Learning Objectives

Learning Intentions

By the end of this lesson, students will be able to:

- Define and explain the **purpose** of network protocols
- Differentiate between **TCP** and **UDP**, identifying their key characteristics and use cases
- Describe the functions of **HTTP, HTTPS, and DHCP**
- Analyse real-world scenarios and identify the relevant network protocols involved

Key Vocabulary

Term	Definition
Protocol	A set of rules that govern how data is transmitted and received over a network
TCP (Transmission Control Protocol)	A reliable, connection-oriented protocol that ensures data is delivered correctly and in order
UDP (User Datagram Protocol)	A fast, connectionless protocol that prioritises speed over reliability
HTTP (Hypertext Transfer Protocol)	A protocol for transferring web pages and resources over the internet
HTTPS (HTTP Secure)	An encrypted version of HTTP that provides secure communication
DHCP (Dynamic Host Configuration Protocol)	A protocol that automatically assigns IP addresses to devices on a network
IP address	A unique identifier for a device on a network

IB ATL Skills

Skill	Description
Communication	Communicating information and ideas effectively
Collaboration	Working collaboratively with peers
Thinking	Applying thinking skills critically and creatively
Information Literacy	Finding and interpreting information

2. What Are Network Protocols?

Definition

> **A network protocol is a set of rules that govern how data is transmitted and received over a network.**

Protocols ensure that devices can communicate with each other, regardless of their manufacturer, operating system, or location. They define:

- **How data is formatted**
- **How data is transmitted**
- **How errors are handled**
- **How connections are established and terminated**

The TCP/IP Model

The TCP/IP model organises network protocols into four layers:

Layer	Function	Protocols
Application Layer	Provides services for user applications	HTTP, HTTPS, FTP, SMTP, DNS, DHCP
Transport Layer	Ensures data transfer between applications	**TCP, UDP**
Network Layer	Routes data between networks	IP
Network Access Layer	Handles physical data transmission	Ethernet, Wi-Fi

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│                    TCP/IP MODEL                                │
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│                                                                 │
│    ┌─────────────────────────────────────────────────────────┐  │
│    │                  APPLICATION LAYER                    │  │
│    │        HTTP • HTTPS • FTP • SMTP • DNS • DHCP        │  │
│    └─────────────────────────────────────────────────────────┘  │
│                              │                                  │
│    ┌─────────────────────────────────────────────────────────┐  │
│    │                  TRANSPORT LAYER                       │  │
│    │                    TCP • UDP                          │  │
│    └─────────────────────────────────────────────────────────┘  │
│                              │                                  │
│    ┌─────────────────────────────────────────────────────────┐  │
│    │                    NETWORK LAYER                        │  │
│    │                       IP                                │  │
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│                              │                                  │
│    ┌─────────────────────────────────────────────────────────┐  │
│    │                 NETWORK ACCESS LAYER                    │  │
│    │                 Ethernet • Wi-Fi                        │  │
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3. Transport Layer Protocols

TCP (Transmission Control Protocol)

Aspect	Description
Type	Connection-oriented
Key Feature	Reliable — guarantees delivery, correct order, and error-free data
How It Works	Establishes a connection (three-way handshake); sends packets; acknowledges receipt; retransmits lost packets
Speed	Slower (overhead from connection establishment and error checking)
Best For	Applications where data integrity is critical
Examples	Web browsing (HTTP/HTTPS), file transfer (FTP), email (SMTP)

Analogy: TCP is like a **registered courier service**—the package is tracked, delivery is confirmed, and everything arrives in the correct order. If something is missing, they send it again.

UDP (User Datagram Protocol)

Aspect	Description
Type	Connectionless
Key Feature	Fast — no connection establishment, no error checking, no retransmission
How It Works	Sends data packets (datagrams) without establishing a connection; no acknowledgment of receipt
Speed	Faster (lower overhead)
Best For	Applications where speed is more important than reliability
Examples	Video streaming, online gaming, VoIP (voice calls), DNS lookups

Analogy: UDP is like **standard post**—the letter is sent, but there's no tracking or guarantee it will arrive. It's fast, but you don't know if it was delivered.

TCP vs. UDP Comparison Table

Feature	TCP	UDP
Connection	Connection-oriented	Connectionless
Reliability	Reliable (guaranteed delivery)	Unreliable (no guarantee)
Ordering	Packets delivered in correct order	Packets may arrive out of order
Error Checking	Yes	No (checksum only)
Retransmission	Lost packets are resent	No retransmission
Speed	Slower	Faster
Overhead	Higher	Lower
Use Cases	Web, email, file transfer	Streaming, gaming, VoIP

Feature	TCP	UDP
Analogy	Registered courier	Standard post

When to Use TCP vs. UDP

Scenario	Best Protocol	Reason
Downloading a file	TCP	File must arrive intact and complete
Loading a web page	TCP	Entire page must be displayed correctly
Sending an email	TCP	Email content must not be lost or corrupted
Streaming a live video	UDP	Speed matters; occasional packet loss is acceptable
Playing an online game	UDP	Low latency is critical; speed over reliability
DNS query	UDP	Small request; speed is more important than reliability
Video call (VoIP)	UDP	Real-time communication; slight audio/video loss is acceptable

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4. Application Layer Protocols

HTTP (Hypertext Transfer Protocol)

Aspect	Description
Definition	A protocol for transferring web pages and resources over the internet
Key Feature	The foundation of the World Wide Web
How It Works	Client (browser) sends a request; server sends a response (web page)
Port	80 (default)
Security	**Not secure** — data is transmitted in plaintext

What happens when you visit a website:

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1. Browser sends HTTP request to server
2. Server processes the request
3. Server sends HTTP response (web page content)
4. Browser displays the content
```

HTTPS (HTTP Secure)

Aspect	Description
Definition	An encrypted version of HTTP that provides secure communication
Key Feature	Uses TLS/SSL encryption to protect data
How It Works	Same as HTTP, but data is encrypted before transmission
Port	443 (default)
Security	**Secure** — data is encrypted (confidentiality, integrity, authentication)

Why HTTPS is Important:

Reason	Description
Encryption	Protects data from eavesdropping
Authentication	Verifies the identity of the server
Integrity	Ensures data is not tampered with during transmission
Data Protection	Secures sensitive information—login credentials, financial data, personal information

Analogy: HTTP is like a **postcard**—anyone can read it along the way. HTTPS is like a **sealed letter**—it's secure and can only be read by the intended recipient.

DHCP (Dynamic Host Configuration Protocol)

Aspect	Description
Definition	A protocol that automatically assigns IP addresses to devices on a network
Key Feature	Automates IP address management; eliminates manual configuration
How It Works	Device requests an IP address; DHCP server assigns one from a pool
Benefits	Simplifies network management; prevents IP address conflicts
When Used	When a device connects to a new network (Wi-Fi, Ethernet)

How DHCP Works:

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┌─────────────────────────────────────────────────────────────────┐
│                 DHCP PROCESS                                    │
├─────────────────────────────────────────────────────────────────┤
│                                                                 │
│   Device:        "I'm new here. Can I have an IP address?"     │
│                     ↓                                           │
│   DHCP Server:   "Here is your IP address: 192.168.1.42."     │
│                   "Your subnet mask is 255.255.255.0."        │
│                   "Your default gateway is 192.168.1.1."      │
│                     ↓                                           │
│   Device:        "Thank you! I'll use this address."           │
│                                                                 │
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What DHCP Provides:

Information	Description
IP Address	Unique address for the device
Subnet Mask	Defines the network segment
Default Gateway	IP address of the router (to reach other networks)
DNS Server	IP address for domain name resolution

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5. Protocol Summary Table

Protocol	Layer	Function	Key Feature	Example Use
TCP	Transport	Reliable data delivery	Connection-oriented	Web, email, file transfer
UDP	Transport	Fast, lightweight data delivery	Connectionless	Streaming, gaming, VoIP
HTTP	Application	Transfer web pages	Unencrypted	Browsing non-secure websites
HTTPS	Application	Secure web transfer	Encrypted	Banking, shopping, login pages
DHCP	Application	Automatic IP assignment	Automates IP management	Joining a Wi-Fi network

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6. Real-World Protocol Scenarios

Scenario	Protocols Involved	Why?
Downloading a file	**TCP, HTTP/HTTPS**	TCP ensures reliability; HTTP/HTTPS transfers the file
Live streaming on YouTube	**UDP, HTTP/HTTPS**	UDP for streaming data; HTTP for control and metadata
Logging into email	**TCP, HTTPS, SMTP/POP3**	TCP for reliability; HTTPS for secure login; SMTP/POP3 for email
Playing online games	**UDP**	Speed matters; occasional packet loss is acceptable
Browsing a shopping site	**TCP, HTTPS**	TCP for reliability; HTTPS for secure transactions
Connecting to Wi-Fi	**DHCP**	Automatically assigns an IP address
Video call (Zoom/Teams)	**UDP (preferred)**	Real-time; speed over reliability
Sending an email	**TCP, SMTP**	TCP ensures delivery; SMTP sends the email

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7. Lesson Sequence

Lesson Plan

Phase	Time	Activity
Engage	5 min	"Internet Blackout" scenario—imagine internet stops working; discuss importance of network communication
Explore	15 min	Review TCP/IP model; "Protocol Charades" — groups act out protocols
Explain	20 min	Transport layer (TCP vs. UDP with analogies); Application layer (HTTP, HTTPS, DHCP); quick quiz
Elaborate	15 min	Real-world scenarios — identify protocols involved and explain their roles
Wrap-up	5 min	Key takeaways; exit ticket

Activity: Protocol Charades

Instructions:

1. Divide students into small groups
2. Each group selects a protocol (TCP, UDP, HTTP, HTTPS, DHCP)
3. Groups act out the function of their protocol **without speaking**
4. Other groups guess which protocol is being demonstrated

Suggested Actions:

- **TCP:** Establish a handshake, send a package, wait for confirmation, send again if lost
- **UDP:** Send a package quickly, move on without waiting for confirmation
- **HTTP:** Type a URL, request a page, receive content in plaintext
- **HTTPS:** Same as HTTP but mime locking/encrypting the data
- **DHCP:** Raise hand to ask for an address, receive an "IP card"

Activity: Real-World Scenario Analysis

Instructions: For each scenario, identify the protocols involved and explain why each is used.

Scenario	Protocols Involved	Justification
Downloading a file	TCP, HTTP/HTTPS	TCP ensures reliable file transfer; HTTP/HTTPS used to request and transfer the file
Watching a live stream on YouTube	UDP, HTTP/HTTPS	UDP for real-time streaming; HTTP/HTTPS for initial request and control information
Logging into school email	TCP, HTTPS, SMTP/POP3	TCP for reliable delivery; HTTPS for secure login; SMTP to send; POP3/IMAP to receive
Playing a multiplayer game	UDP	Speed is critical; occasional packet loss is acceptable

Scenario	Protocols Involved	Justification
Connecting to school Wi-Fi	DHCP	Automatically assigns IP address to device

8. Assessment Activities

Assessment Type	Description
Class Participation	Observe engagement and contributions during discussions
Protocol Charades	Assess understanding of protocol functions through performances
Quick Quiz	Use online tools (Quizlet) to assess understanding of protocol characteristics
Real-World Scenarios	Assess ability to apply knowledge to identify protocols in different situations
Exit Ticket	Gauge understanding and identify areas needing clarification

9. Differentiation

Level	Strategy
Support	Provide visual aids, concise definitions, and real-world examples
Challenge	Encourage advanced students to research more complex protocols or investigate network security issues related to protocols

10. Resources

- **Presentation** (Slide Deck)
- **eBook** (A2.1.4 with quiz)
- **Worksheet:** Protocol comparison + scenario questions
- **Quizlet** (A2.1.4 vocabulary flashcards)

11. Summary: Key Takeaways

Protocol	Key Point
TCP	Reliable, connection-oriented; used for web, email, file transfer
UDP	Fast, connectionless; used for streaming, gaming, VoIP
HTTP	Unencrypted web transfer; used for standard websites
HTTPS	Encrypted web transfer; used for secure websites (banking, shopping)
DHCP	Automatically assigns IP addresses; used when connecting to networks

> *"Network protocols are the languages that enable devices to communicate. Understanding them helps us see how data flows seamlessly across the internet."*